

Project Documentation: IMDB Sentiment Analysis with PyTorch

1. Data Pipeline (data.py)

- Text Preprocessing: Converts text to lowercase, removes HTML tags, expands contractions.
- Vocabulary Management: Limits vocabulary to 25,000 words. Handles <pad> and <unk> tokens.
- Dataset & DataLoader: Loads .txt files, pads sequences, and sorts them by length for efficient RNN processing.

2. Neural Network Architecture (model.py)

- Embedding Layer: Converts token indices into dense word vectors.
- Recurrent Layer: Configurable via `cell_type` ('lstm' or 'rnn'). Uses `pack_padded_sequence` to ignore <pad> tokens.
- Fully Connected Layer: Maps final hidden state to 2 classes (positive/negative).

3. Execution and Training (main.py)

- Hyperparameters: Configured at the top (Batch Size: 64, Embed Size: 100, etc.).
- Training Loop: Uses `CrossEntropyLoss` and Adam optimizer.
- Evaluation Loop: Runs `model.eval()` to calculate final accuracy on the test set.

How to Run:

1. Ensure the IMDB dataset path matches `BASE_DATA_PATH`.
2. Set `cell_type='lstm'` or `cell_type='rnn'` in `main.py`.
3. Run: `python main.py ""`